

Error Analysis: Find the inverse of $f(x) = 5^{x-6} + 2$

$$y = 5^{x-6} + 2$$

Write in $y = f(x)$ form.

$$x = 5^{y-6} + 2$$

Interchange x and y .

$$x - 2 = 5^{y-6}$$

Subtract 2 from each side.

$$y - 6 = \log_5 x - 2$$

Rewrite in logarithmic form.

$$y = \log_5 x - 2 + 6$$

Add 6 to each side.

$$y = \log_5 x + 4$$

Simplify.

$$f^{-1}(x) = \log_5 x + 4$$